

Semester: Aug 2025 – Nov 2025		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1 Hr 15 Min
MSE		
Programme code: Programme: B.Tech	Class: FY	Semester: I (SVU R-2025)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH
Course Code: 216U06C101	Name of the Course: Applied Mathematics-I	

Question No.		Max. Marks
Q:1	Reduce the matrix $A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 2 \\ 7 & 4 & 10 \\ 8 & 5 & 8 \end{bmatrix}$ to Normal form and hence find the rank	06
Q:2	Attempt any TWO from the following.	12
1	If $u = x(1 - y)$, $v = xy(1 - z)$, $w = xyz$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$	06
2	Encrypt and decrypt the message "I AM FINE " using the orthogonal matrix $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	06
3	If $u = x^2y + y^2z + z^2x$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2(x + y + z)$	06
Q:3	Attempt any TWO from the following.	12
1	Use Gauss-Seidel method to solve the following equations (Take three iterations) $15x + y + z = 17$ $2x + 15y + z = 18$ $x + 2y + 15z = 18$	06
2	For which values of λ following set of equations is consistent? Find and solve equations for those values $x + 2y + z = 3$, $x + y + z = \lambda$, $3x + y + 3z = \lambda^2$	06
3	If $u = x^3 \sin^{-1} \left(\frac{\sqrt{y} + \sqrt{x}}{\sqrt{y} - \sqrt{x}} \right)$ then prove that i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$ ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 6u$	06

SOLUTION SCHEME

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Question No.		Max. Marks
Q:1	Reduce the matrix $A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 2 \\ 7 & 4 & 10 \\ 8 & 5 & 8 \end{bmatrix}$ to Normal form and hence find the rank Solution: $\begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 2 \\ 7 & 4 & 10 \\ 8 & 5 & 8 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & -2 \\ 3 & 2 & 2 \\ 7 & 4 & 10 \\ 8 & 5 & 8 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 7R_1 \\ R_4 \rightarrow R_4 - 8R_1 \end{array}$ $\rightarrow \begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 8 \\ 0 & -3 & 24 \\ 0 & -3 & 24 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 3R_2 \\ R_4 \rightarrow R_4 - 3R_2 \end{array} \begin{bmatrix} 1 & 1 & -2 \\ 0 & -1 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 + 2C_1 \end{array}$ $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{C_2 \rightarrow -C_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{C_3 \rightarrow C_3 - 8C_2}$ $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$ $\Rightarrow \rho(A) = 2$	03 06
Q:2	Attempt any TWO from the following.	12

1	If $u = x(1 - y)$, $v = xy(1 - z)$, $w = xyz$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ Solution: $\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 - y & -x & 0 \\ y(1 - z) & x(1 - z) & -xy \\ yz & zx & xy \end{vmatrix}$ $= (1 - y)[(x - xz)xy + xy \cdot zx] + x[(y - yz)xy + xy \cdot yz]$ $= (1 - y)[x^2y - x^2yz + x^2yz] + x[xy^2 - xy^2z + xy^2]$ $= (1 - y)(x^2y) + x(xy^2)$ $= x^2y - x^2y^2 + x^2y^2 = x^2y$	03 06																																													
2	Encrypt and decrypt the message "I AM FINE " using the orthogonal matrix $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ Solution: Encryption (encode the message) Convert the message into a numeric format .Let us use 1-based system, where <table><tr><td>Letter</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td></td><td>H</td><td>...</td><td>L</td><td>....</td><td>O</td><td>....</td><td>Z</td><td>space</td></tr><tr><td>Value</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td></td><td>8</td><td>....</td><td>1</td><td>...</td><td>1</td><td>....</td><td>2</td><td>0 or</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>.</td><td>2</td><td>...</td><td>5</td><td>.</td><td>6</td><td>27</td></tr></table> Split into 2-character blocks Block 1: "I " $\rightarrow [9, 0]$; Block 2: "AM" $\rightarrow [1, 13]$;Block 3: " F " $\rightarrow [0, 6]$ Block 4: "IN" $\rightarrow [9, 14]$;Block 5: " E " $\rightarrow [5, 0]$ Arrange the numbers into a matrix $M = \begin{pmatrix} 9 & 1 & 0 & 9 & 5 \\ 0 & 13 & 6 & 14 & 0 \end{pmatrix}$ $C = Q.M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 9 & 1 & 0 & 9 & 5 \\ 0 & 13 & 6 & 14 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -13 & -6 & -14 & 0 \\ 9 & 1 & 0 & 9 & 5 \end{pmatrix}$ The message is transmitted into the linear form as 0,9,-13,1,-6,0,-14,9,0,5 Decryption Since $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is orthogonal, $Q^{-1} = Q^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ $M = Q^T.C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -13 & -6 & -14 & 0 \\ 9 & 1 & 0 & 9 & 5 \end{pmatrix} = \begin{pmatrix} 9 & 1 & 0 & 9 & 5 \\ 0 & 13 & 6 & 14 & 0 \end{pmatrix}$ The columns of the above matrix are written in linear form as 9,0,1,13,0,6,9,14,5,0 Since 9 is assigned to I, 0 to space, and 5 to E, the decoded message is "I AM FINE ".	Letter	A	B	C	D	E		H	...	L		O		Z	space	Value	1	2	3	4	5		8		1	...	1		2	0 or									.	2	...	5	.	6	27	04 06
Letter	A	B	C	D	E		H	...	L		O		Z	space																																	
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3	If $u = x^2y + y^2z + z^2x$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2(x + y + z)$ Solution:																																														

	$\frac{\partial u}{\partial x} = 2xy + z^2, \quad \frac{\partial u}{\partial y} = x^2 + 2yz, \quad \frac{\partial u}{\partial z} = y^2 + 2zx$ $\frac{\partial^2 u}{\partial x^2} = 2y, \quad \frac{\partial^2 u}{\partial y^2} = 2z, \quad \frac{\partial^2 u}{\partial z^2} = 2x$ $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2y + 2z + 2x = 2(x + y + z)$	04 06
Q:3	Attempt any TWO from the following.	12
1	Use Gauss-Seidel method to solve the following equations (Take three iterations) $15x + y + z = 17$ $2x + 15y + z = 18$ $x + 2y + 15z = 18$ Solution: $x = \frac{1}{15}(17 - y - z), \quad y = \frac{1}{15}(18 - 2x - z), \quad z = \frac{1}{15}(18 - x - 2y)$ First Iteration put $y=0, z=0$ $x = \frac{17}{15} = 1.133$ $y = \frac{1}{15}(18 - 2 \times 1.133 - 0) = 1.049$ $z = \frac{1}{15}(18 - 1.133 - 2 \times 1.049) = 0.985$ Second Iteration $x = \frac{1}{15}(17 - 1.049 - 0.985) = 0.998$ $y = \frac{1}{15}(18 - 2 \times 0.998 - 0.985) = 1.001$ $z = \frac{1}{15}(18 - 0.998 - 2 \times 1.001) = 1.000$ Third Iteration $x = \frac{1}{15}(17 - 1.001 - 1.000) = 1.000$ $y = \frac{1}{15}(18 - 2 \times 1.000 - 1.000) = 1.000$ $z = \frac{1}{15}(18 - 1.000 - 2 \times 1.000) = 1.000$	03 06
2	For which values of λ following set of equations is consistent? Find and solve equations for those values $x + 2y + z = 3, \quad x + y + z = \lambda, \quad 3x + y + 3z = \lambda^2$ Solution: consider $[A:B] = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 1 & 1 & 1 & \lambda \\ 3 & 1 & 3 & \lambda^2 \end{bmatrix}$ Applying $R_2 - R_1$ and $R_3 - 3R_1$, we get $[A:B] \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & \lambda - 3 \\ 0 & -5 & 0 & \lambda^2 - 9 \end{bmatrix}$	

	Applying $R_3 - 5R_2$, we get $[A:B] \sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & \lambda - 3 \\ 0 & 0 & 0 & \lambda^2 - 5\lambda + 6 \end{bmatrix}$ For consistency of the equation, the rank of A and rank of $[A:B]$ must be the same. From above reduced form of $[A:B]$ it is clear that the rank of $A = 2$. To have the rank of $[A:B] = 2$, Consider $\lambda^2 - 5\lambda + 6 = 0$ i. e., $(\lambda - 2)(\lambda - 3) = 0 \Rightarrow \lambda = 2$ and $\lambda = 3$ (i) Solution for $\lambda = 2$ The reduced Echelon form of $[A:B]$ is $\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & \lambda - 3 \\ 0 & 0 & 0 & \lambda^2 - 5\lambda + 6 \end{bmatrix}$ Substituting $\lambda = 2$, we have $\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ The reduced form of linear equations is $x + 2y + z = 3$ and $y = 1$ Let $z = k$, an arbitrary constant, $\therefore x = 1 - k$ Hence $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 - k \\ 1 \\ k \end{bmatrix}$ has infinite values as k varies (ii) Solution for $\lambda = 3$: The reduced Echelon form of $[A:B]$ By substituting $\lambda = 3$, we have $\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ The reduced form of linear equations is $x + 2y + z = 3$ and $y = 0$ Let $z = c$, an arbitrary constant. $\therefore x = 3 - c$ Hence $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 - c \\ 0 \\ c \end{bmatrix}$ has infinite solutions as c varies.	02 04 06
3	If $u = x^3 \sin^{-1} \left(\frac{\sqrt{y} + \sqrt{x}}{\sqrt{y} - \sqrt{x}} \right)$ then prove that i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$ ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 6u$ Solution:	

Replace x by xt and $y = yt$ to get

$$u = (xt)^3 \sin^{-1} \left(\frac{\sqrt{yt} + \sqrt{xt}}{\sqrt{yt} - \sqrt{xt}} \right)$$

$$= t^3 x^3 \sin^{-1} \left(\frac{\sqrt{y} + \sqrt{x}}{\sqrt{y} - \sqrt{x}} \right)$$

$\Rightarrow u$ is homogeneous function of degree 3 in var. x and y

By Euler's Thm " If z is homogeneous function of two variables x and y of degree n , then

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z "$$

$$\text{and corl " } x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n(n-1) z "$$

$$(i) x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = 3u$$

and

$$(ii) x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u = 3 \times 2u = 6u$$

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